

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA",MACHHE, BELAGAVI-590018



Activity Report Cloud Data Engineering On Apache Nifi

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Manoj Preetham V K (1CD23AI060)

Varshini A (1CD23AI120)

Abhishek Karyagol (1CD23AI007)

Soujanya K (1CD23AI103)

**Under the Guidance
of**

Dr. Priyanka R

Assistant Professor,
Department of AIML, CITech.



**Department of Artificial Intelligence and Machine Learning
CAMBRIDGE INSTITUTE OF TECHNOLOGY, BANGALORE-560 036
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INTRODUCTION

Cloud Data Engineering focuses on designing, building, and managing **scalable data pipelines** using cloud-based infrastructure. With the rapid growth of data from applications, IoT devices, and digital platforms, organizations require systems that can handle **high-volume, high-velocity, and diverse data sources**.

Apache NiFi plays a key role in this domain by enabling **automated data flow management** across distributed systems. It supports seamless data ingestion, transformation, and delivery while ensuring **reliability and fault tolerance**.

Apache NiFi is widely used in modern cloud architectures to build **real-time data pipelines**, making it an essential tool in cloud-based data engineering solutions.

Purpose:

The purpose of Apache NiFi in Cloud Data Engineering is to:

- Automate **data ingestion from multiple cloud and on-premise sources**
- Enable **real-time and batch data processing**
- Ensure **secure and reliable data transfer across distributed systems**
- Simplify the design of **data pipelines using visual workflows**
- Integrate with cloud platforms and big data ecosystems

It helps organizations efficiently manage data pipelines without requiring extensive manual coding.

Features of Apache NIFI:-

- Drag-and-drop UI for building data pipelines
- Flow-based programming model
- Real-time data ingestion and streaming support

- Back pressure mechanism for handling large-scale data
- Data provenance for tracking data lifecycle
- Fault tolerance and guaranteed delivery
- Secure data transmission using SSL and authentication
- Horizontal scalability through clustering
- Integration with cloud services and APIs

STEP BY STEP WORKFLOW:

- Ingest data from cloud sources (APIs, cloud storage, databases)
- Buffer data using internal queues
- Preprocess data (cleaning, filtering, formatting)
- Transform data into required formats
- Route data dynamically based on conditions
- Deliver data to cloud destinations (data lakes, warehouses, analytics tools)
- Monitor pipeline performance in real-time

Real World Applications:

- **Real-Time Data Pipelines:** Streaming data to cloud analytics platforms
- **Cloud Data Migration:** Moving data between on-premise and cloud systems
- **IoT Data Processing:** Handling large-scale sensor data in cloud
- **Log and Event Processing:** Centralized logging in cloud environments
- **Data Lake Integration:** Feeding data into cloud storage systems

Installation Steps:

1. Download Apache NiFi from the official website
2. Deploy on:
Cloud Virtual Machines (AWS EC2, Azure VM, GCP Compute Engine)
3. Configure cluster setup for scalability
4. Start NiFi services
5. Access UI via browser
6. Integrate with cloud storage and APIs

PROS & CONS :

Pros:

- Scalable for cloud-based architectures
- Supports real-time streaming pipelines
- Easy-to-use visual interface
- Strong fault tolerance and reliability
- Seamless integration with cloud and big data tools
- Reduces development time

Cons:

- High resource consumption in cloud environments
- Requires careful cluster configuration
- Not suitable for heavy data analytics
- Workflow complexity increases at scale
- Mapping to DMDW Syllabus

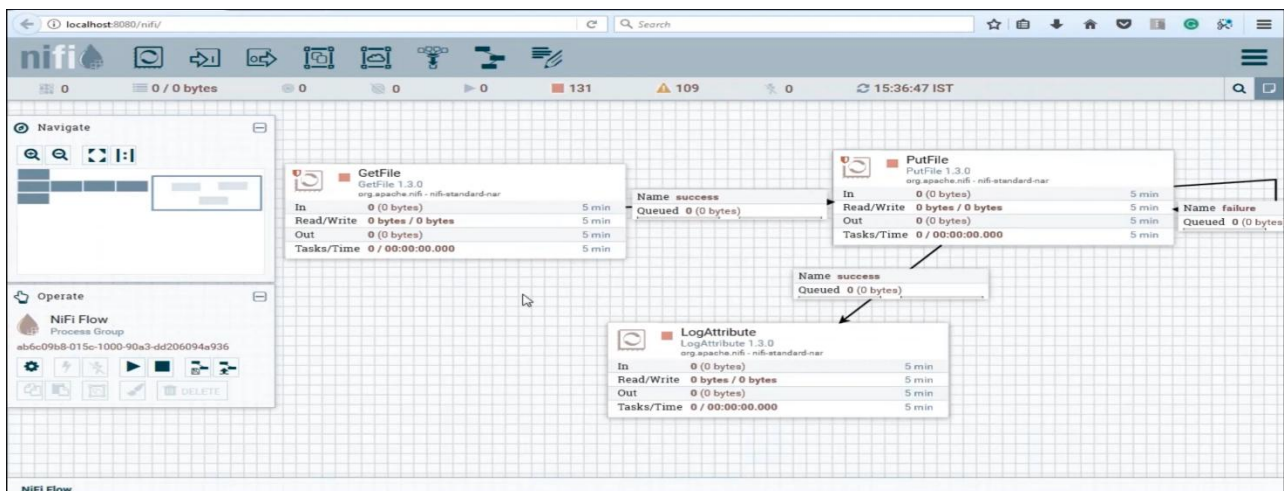
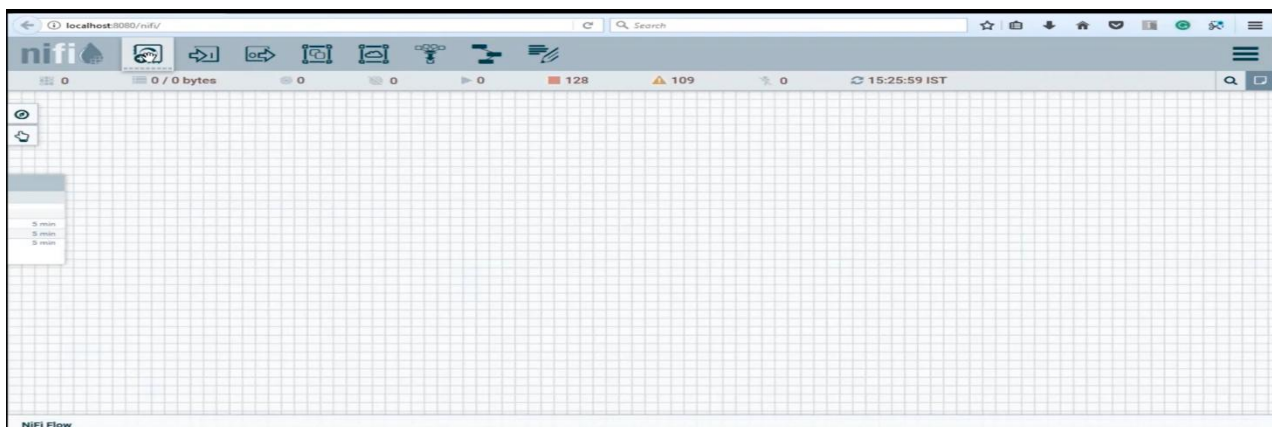
ROLE IN CLOUD DATA ENGINEERING:

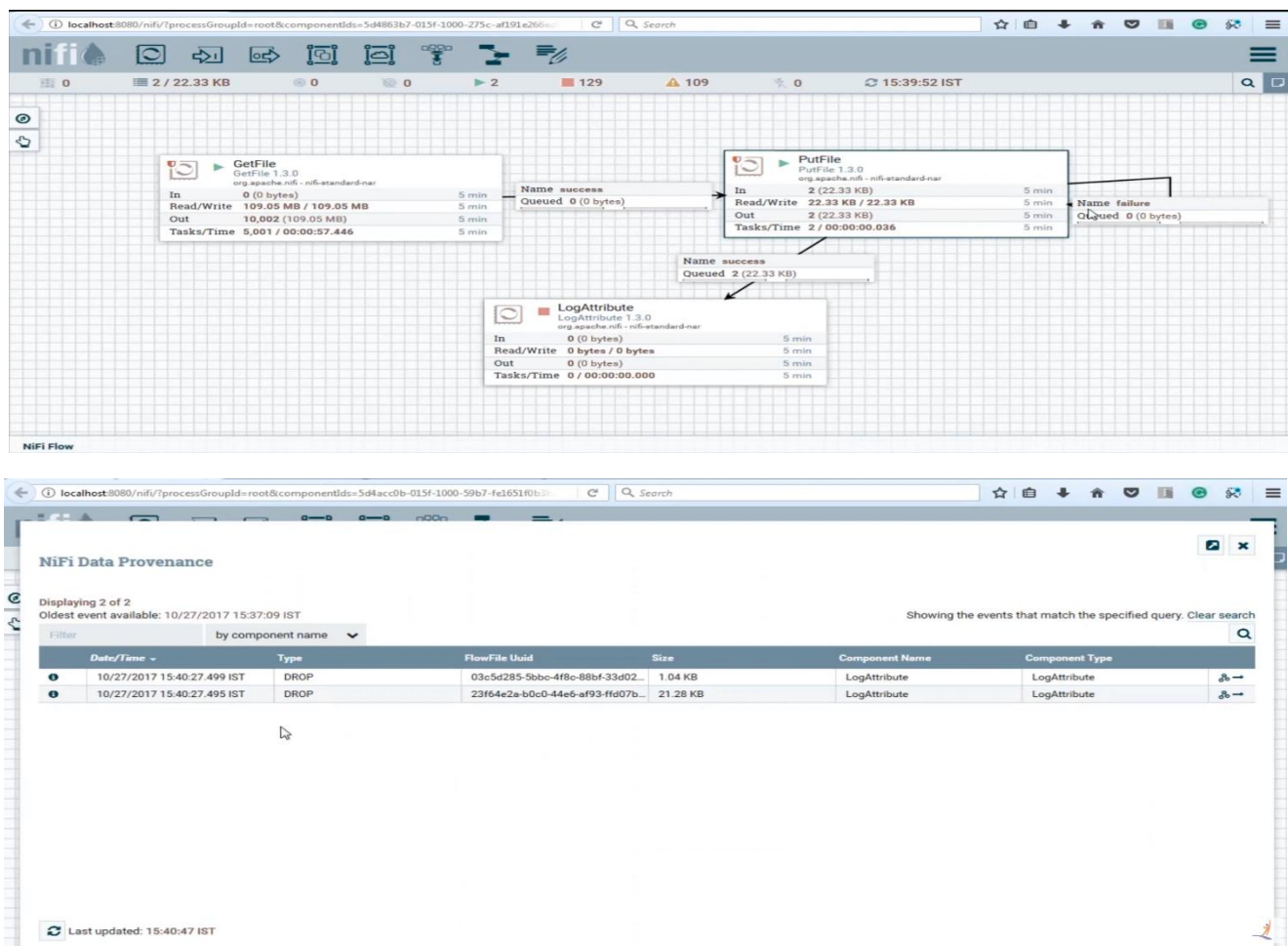
Apache NiFi plays a crucial role in:

- Building **ETL/ELT pipelines in cloud**
- Enabling **real-time data streaming architectures**
- Acting as a **data ingestion layer for data lakes**
- Supporting **hybrid cloud data integration**
- Ensuring **data governance through provenance tracking**

It acts as a bridge between:

- Data sources
- Processing system
- Cloud storage and analytics platforms





CONCLUSION

Apache NiFi is a powerful tool for building **scalable, reliable, and real-time data pipelines** in cloud environments. Its visual programming model simplifies pipeline development, while its robust architecture ensures efficient data handling across distributed systems.

In Cloud Data Engineering, NiFi serves as a critical component for **data ingestion, transformation, and integration**, making it highly valuable for modern data-driven applications.

